AMARNATH K R

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SUMMARY

Entry-level PCB design engineer (B.Tech ECE) with four board designs in Altium Designer and KiCad, from schematic and component calculations to DRC-clean layout and Gerber output. Experience with mains-powered safety layouts, linear power supplies, and 2- and 4-layer boards.

SKILLS

PCB Design	Schematic Capture, 2 & 4-Layer Layout, Footprint & Library Management, Ground Pours & Stitching Vias, HV/SELV Creepage & Clearance, ERC & DRC, Gerber & Drill Generation
EDA Tools	Altium Designer, KiCad, Cadence Virtuoso, Proteus
Circuit Design	Linear Power Supplies, Current Limiting, 555 Timer Circuits, Transistor/Relay Drivers, Isolated AC-DC Supply, Component Calculations
Lab & Debug	Oscilloscope, Multimeter, Bench Power Supply, Function Generator, Soldering, Hardware Debugging
Hardware Design	ESP32, STM32, Arduino, Embedded C, MicroPython, UART, SPI, I2C

PROJECTS

PIR Proxi Machine Cut-Off Circuit – 2-Layer PCB, Altium Designer *(2026)* GitHub

- Designed a mains-powered safety interlock PCB using a PIR sensor, LM555 monostable timer, transistor relay driver, and isolated AC-DC power supply.
- Applied high-voltage/SELV creepage and clearance rules, HV ground-pour exclusion, and dedicated safety markings.

Edge Anomaly Detection Board – 4-Layer PCB, KiCad *(2026)* GitHub

- Designed a 4-layer PCB around the ESP-12E integrating a DHT11 sensor, I2C display, UART debug header, and external alert output.
- Completed schematic capture, component interfacing, layout, DRC verification, and 3D review.

LM317 Adjustable Power Supply – 2-Layer PCB, KiCad *(2026)* GitHub

- Designed an adjustable linear supply with LM358/BC547 constant-current limiting to protect both regulator and load.
- Completed 2-layer layout with ground planes and stitching vias; passed ERC/DRC and generated fabrication-ready Gerber and drill files.

Environmental Monitoring System – 2-Layer PCB, KiCad *(2026)* GitHub

- Designed a 2-layer ESP32 breakout PCB with CH340G USB-to-UART, AMS1117-3.3 regulation, DHT11 footprint, and GND copper pour; exported Gerbers.

2.4 GHz RF Front-End Transmitter – Cadence Virtuoso *(2026)* GitHub

- Designed a Gilbert-cell mixer, LC-VCO, and Class-A PA; validated post-layout with Calibre PEX (22.7% transmitter efficiency at 0.9 V).

EXPERIENCE

Apprentice – ERCMPU Ltd, Thrissur Dairy Jul 2019 – Jul 2020

- Maintained industrial control panels and pump systems; diagnosed faults and supported preventive maintenance.

Trainee – FCI-OEN Connectors Pvt Ltd, Ernakulam Jan 2019 – Jul 2019

- Inspected and tested connector assemblies on a high-volume electronics line; reported defects for quality control.

EDUCATION

B.Tech, Electronics & Communication Engg. – GEC Thrissur *(CGPA 7.46/10)* Aug 2023 – May 2026

Diploma, Electronics Engg. – SRGPTC Thriprayar *(CGPA 8.59/10)* Jan 2020 – May 2022

Earlier: ITI Electronics Mechanic (2018) – 73% | Higher Secondary (2016) – 69% | SSLC (2014) – 81%

CERTIFICATIONS

Electronics System Design: Hands-on Circuit and PCB Design – IIT Delhi (NPTEL) Nov 2026

Honors Diploma in Computer Hardware and Networking – GTEC Education Jul 2024

National Apprenticeship Certificate – Directorate General of Training (DGT) May 2022

AI Developer Professional Certificate – IBM (via Coursera) Nov 2025